

Mechanistic Fixed-Bed CO₂ Adsorption Model

Part A — Derivation from Conservation Laws

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A.1 Gas-phase CO₂ mass balance

Take a control volume $[z, z + \Delta z]$ of cross-section A (bed basis). Moles of gaseous CO₂ inside: $\varepsilon c A \Delta z$.

Fluxes through the faces (per bed area, mol m⁻² s⁻¹):

$$N(z, t) = \underbrace{uc}_{\text{convective}} - \underbrace{\varepsilon D_L \frac{\partial c}{\partial z}}_{\text{dispersive (Fickian)}}.$$

The convective term is $uc = \varepsilon v_i c$: interstitial molar flux $v_i c$ scaled by the open-area fraction ε . The dispersive flux acts on the same open area, hence the ε .

Sink to the solid. Sorbent mass in the CV is $(1 - \varepsilon)\rho_p A \Delta z$, so uptake removes $(1 - \varepsilon)\rho_p \frac{\partial q}{\partial t} A \Delta z$ mol s⁻¹ from the gas.

Balance (accumulation = in – out – sink), divide by $A \Delta z$, let $\Delta z \rightarrow 0$:

$$\varepsilon \frac{\partial c}{\partial t} + \frac{\partial N}{\partial z} = -(1 - \varepsilon)\rho_p \frac{\partial q}{\partial t},$$

and with A2 ($u_z = 0$):

$$\boxed{\varepsilon \frac{\partial c}{\partial t} + u \frac{\partial c}{\partial z} = \varepsilon D_L \frac{\partial^2 c}{\partial z^2} - (1 - \varepsilon)\rho_p \frac{\partial q}{\partial t}} \quad (\text{A.1})$$

Term	Physical meaning	Unit
εc_t	accumulation of gaseous CO ₂ per bed volume (ε converts gas-basis c to bed basis)	mol m ⁻³ s ⁻¹
$u c_z$	net convective outflow per bed volume, superficial u	mol m ⁻³ s ⁻¹
$\varepsilon D_L c_{zz}$	net dispersive inflow (axial mixing)	mol m ⁻³ s ⁻¹
$(1 - \varepsilon)\rho_p q_t = \alpha_b q_t$	transfer to the adsorbed phase	mol m ⁻³ s ⁻¹

Remark A.1 (variable velocity — when 10–15 % feeds bite). Only CO₂ leaves the gas; a total molar balance with $c_{\text{tot}} = P/RT$ constant (A3, isothermal) gives

$$c_{\text{tot}} \frac{\partial u}{\partial z} = -(1 - \varepsilon)\rho_p \frac{\partial q}{\partial t} \implies u(z, t) = u_{\text{in}} - \frac{\alpha_b}{c_{\text{tot}}} \int_0^z q_t dz'.$$

Across a saturating front the velocity deficit is $\Delta u/u \approx y_f$: $\sim 1\%$ at 1% feed (negligible) but $\sim 10\text{--}15\%$ for runs 4–8 (**new runs/**). The dilute model (??) then mislocates t_{st} by $O(y_f)$; if that exceeds the $\pm 10\%$ Gate-B budget, solve (??) together with the u -equation above (still MOL-compatible: u is obtained by quadrature of q_t at each RHS evaluation).

A.2 Solid-phase balance: LDF kinetics

Exact pellet-scale mass transfer is a diffusion problem in the particle; the **linear driving force** closure replaces it by first-order relaxation toward equilibrium:

$$\boxed{\frac{\partial q}{\partial t} = k (q^*(c, T) - q)} \quad (\text{A.2})$$

Term	Physical meaning	Units
q_t	uptake rate per kg sorbent	$\text{mol kg}^{-1} \text{s}^{-1}$
k	inverse relaxation time of the pellet	s^{-1}
$q^* - q$	displacement from equilibrium (driving force)	mol kg^{-1}

Mechanistic content of k (resistances in series, Glueckauf; Ruthven ch. 6): for spherical pellets of radius r_p , film coefficient k_f , macropore diffusivity D_p , porosity ε_p , crystal/micro scale r_c , D_c , and local slope $K' = \rho_p \partial q^* / \partial c$,

$$\frac{1}{k} = \frac{r_p K'}{3k_f} + \frac{r_p^2 K'}{15 \varepsilon_p D_p} + \frac{r_c^2}{15 D_c},$$

each term a time constant (s). LDF is exact for a linear isotherm with parabolic intraparticle profile; otherwise it is the controlled approximation this project fits at Gate B. For amine sorbents (PEI@SiO₂) the last, reaction/amine-phase term often dominates and k is small — the measured, strongly smeared fronts (**breakthrough.out/run 5**: $t_b \approx 14$ s vs $t_{50} \approx 230$ s) are consistent with a kinetics-limited bed (§C.4).

A.3 Equilibrium closure

Toth (primary; amine-functionalised sorbent). Written in the concentration basis,

$$\boxed{q^*(c, T) = \frac{n_s(T) b(T) c}{[1 + (b(T) c)^{t_T}]^{1/t_T}}} \quad (\text{A.3})$$

$$n_s(T) = n_{s0} \exp\left[\chi \left(1 - \frac{T}{T_0}\right)\right], \quad b(T) = b_0 \exp\left[\frac{\Delta H_0}{RT_0} \left(\frac{T_0}{T} - 1\right)\right], \quad t_T(T) = t_0 + \alpha_T \left(1 - \frac{T_0}{T}\right),$$

with $0 < t_T \leq 1$ the heterogeneity exponent ($t_T = 1$ recovers Langmuir). **Units trap:** literature Toth parameters (e.g. the Stampi-Bombelli 2024 set, still ?? in `CLAUDE.md`) are usually *pressure-basis*, b_P in kPa^{-1} with $q^*(p_{\text{CO}_2})$; convert with $p_{\text{CO}_2} = cRT$, i.e. $b(T) = b_P(T) RT$. The conversion inserts an extra T -dependence — do it before differentiating, not after.

Equivalent driving-force form used in `derivation.md` §1.2 ($C - C^*$ with C^* the isotherm inverse): identical model when $k_a a_p (1 - \varepsilon)(C - C^*) = \alpha_b k (q^* - q)$ under the local linearisation $k = k_a a_p (C - C^*) / (\rho_p (q^* - q))$; (??) is preferred because $q^*(c, T)$ stays single-valued while the inverse $C^*(q, T)$ need not be evaluated.

Dual-Site Langmuir (alternative; zeolite 13X):

$$q^*(c, T) = \frac{q_1 b(T) c}{1 + b(T) c} + \frac{q_2 d(T) c}{1 + d(T) c}, \quad b = b_0 e^{-\Delta U_b / RT}, \quad d = d_0 e^{-\Delta U_d / RT}, \quad \Delta U_b, \Delta U_d < 0. \quad (\text{A.4})$$

Lemma A.1 (shape of the closures). *For fixed T and $c > 0$, both (??) and (??) satisfy*

1. $q^*(0, T) = 0$ and Henry limit $q^* \sim n_s b c$ (Toth) resp. $(q_1 b + q_2 d) c$ (DSL);

2. *strict monotonicity* — *Toth*: $\frac{\partial q^*}{\partial c} = \frac{n_s b}{[1 + (bc)^{t_T}]^{(1+t_T)/t_T}} > 0$;

3. *strict concavity* — *Toth*: $\frac{\partial^2 q^*}{\partial c^2} = -n_s b^2 (1 + t_T) (bc)^{t_T-1} [1 + (bc)^{t_T}]^{-(1+2t_T)/t_T} < 0$; *DSL*:
sum of Langmuir terms with $q_i b^2 \cdot (-2)/(1 + bc)^3 < 0$;

4. *saturation* $q^* \rightarrow n_s$ (*Toth*) resp. $q_1 + q_2$ (*DSL*) as $c \rightarrow \infty$.

Proof. Differentiate (??) with $s = (bc)^{t_T}$: $q^* = n_s b c (1 + s)^{-1/t_T}$, $\partial_c q^* = n_s b (1 + s)^{-1/t_T} - n_s b s (1 + s)^{-1/t_T-1} = n_s b (1 + s)^{-(1+t_T)/t_T}$; differentiate again using $\partial_c s = t_T s/c$. $\square$

Concavity = **favorable** isotherm; it is what makes adsorption fronts self-sharpening (Part D). Note for $t_T < 1$ the curvature blows up like c^{t_T-1} as $c \rightarrow 0^+$: q^* is C^1 but not C^2 at $c = 0$ — harmless for well-posedness (the slope stays bounded by $n_s b$) but it steepens the leading foot of the front.

A.4.3 Thermodynamic consistency of $(-\Delta H)$. The isosteric heat implied by the closure is $(-\Delta H_{\text{iso}}) = RT^2 \partial_T \ln p|_{q^*}$. For (??) with $\chi = \alpha_T = 0$ this returns exactly $-\Delta H_{\text{iso}} = -\Delta H_0$, constant; with $\chi \neq 0$ it becomes loading-dependent. Assumption A7: use that same function in (??). Using an isotherm fitted with one ΔH_0 and an energy balance with an unrelated $(-\Delta H)$ violates the Clausius–Clapeyron relation and produces spurious heat.

A.4 Pseudo-homogeneous energy balance

Same control volume; energy carried by gas convection, conducted axially through the composite bed, released by adsorption, exchanged with the wall. Accumulation uses both phases (single T , A6):

$$C_h \frac{\partial T}{\partial t} + u \rho_g c_{p,g} \frac{\partial T}{\partial z} = \lambda_{\text{eff}} \frac{\partial^2 T}{\partial z^2} + (1 - \varepsilon) \rho_p (-\Delta H) \frac{\partial q}{\partial t} - \frac{4h_w}{d_{\text{col}}} (T - T_{\text{wall}}) \quad (\text{A.5})$$

$$C_h = \varepsilon \rho_g c_{p,g} + (1 - \varepsilon) \rho_p c_{p,s} \quad [\text{J m}^{-3} \text{K}^{-1}].$$

Term	Physical meaning	Units
$C_h T_t$	sensible-heat accumulation of gas + solid per bed volume	W m^{-3}
$u \rho_g c_{p,g} T_z$	convected enthalpy gradient (superficial flux $u \rho_g c_{p,g}$, $\text{J m}^{-2} \text{s}^{-1} \text{K}^{-1}$)	W m^{-3}
$\lambda_{\text{eff}} T_{zz}$	effective axial conduction/thermal dispersion	W m^{-3}
$\alpha_b (-\Delta H) q_t$	adsorption heat source; > 0 during uptake since $(-\Delta H) > 0$, $q_t > 0$	W m^{-3}
$\frac{4h_w}{d_{\text{col}}} (T - T_{\text{wall}})$	wall loss; $4/d_{\text{col}}$ = perimeter/area of the cylinder	W m^{-3}

Neglected knowingly: adsorbed-phase heat capacity $(1 - \varepsilon) \rho_p q c_{p,a}$ ($\leq$ a few % of C_h at $q \lesssim 1 \text{ mol kg}^{-1}$), pressure work, kinetic energy. **Adiabatic column:** $h_w = 0$.

Isothermal reduction — quantitative justification (prompt item 3). Two independent smallness arguments:

1. *Heat-release vs wall-loss timescales.* Heat is generated over the front-passage time $\sim t_{st}$ and removed on $\tau_w = C_h d_{\text{col}} / (4h_w)$. The quasi-steady excursion is $\Delta T \approx \Delta T_{\text{ad}} \cdot \tau_w / t_{st}$ where $\Delta T_{\text{ad}} = \alpha_b (-\Delta H) q_f / C_h$ is the adiabatic rise. Bench rig estimate ($d_{\text{col}} = 8.5 \text{ mm}$, $C_h \sim 6.6 \times 10^5 \text{ J m}^{-3} \text{K}^{-1}$, $h_w \sim 30 \text{ W m}^{-2} \text{K}^{-1}$, $t_{st} \sim 600 \text{ s}$): $\tau_w \approx 47 \text{ s}$, $\Delta T_{\text{ad}} \approx 40 \text{ K}$, so $\Delta T \approx 3\text{--}4 \text{ K}$ — mild, and the isothermal model is a defensible base case **for this rig**. For a pilot column ($d_{\text{col}} \gtrsim 5 \text{ cm}$) τ_w grows linearly in d_{col} and the full (??) is mandatory.

2. *Isotherm sensitivity.* The perturbation enters through $b(T)$: $\delta q^*/q^* \sim (\Delta H_0/RT^2) \Delta T \approx (70\,000/8.314/298^2) \times 3.5 \approx 0.33 \dots$ i.e. even a 3–4 K excursion shifts local capacity by $\sim 30\%$ near the front. Conclusion: keep (??) in the model; drop it only after checking *both* numbers for the case at hand. (This is why the model here is non-isothermal by default and the isothermal system is treated as a limit in Part D.)

A.5 Initial and boundary conditions

Clean bed:

$$c(z, 0) = 0, \quad q(z, 0) = 0, \quad T(z, 0) = T_0. \quad (\text{A.6})$$

Inlet ($z = 0$), feed step at $t > 0$. Two admissible choices:

- *Dirichlet (hyperbolic-limit shortcut):* $c(0, t) = c_f$, $T(0, t) = T_f$.
- *Danckwerts flux continuity (correct with dispersion):* upstream of the packing there is no dispersion, so the total flux uc_f must equal the total flux just inside:

$$uc_f = u c(0^+, t) - \varepsilon D_L c_z(0^+, t), \quad u \rho_g c_{p,g} T_f = u \rho_g c_{p,g} T(0^+, t) - \lambda_{\text{eff}} T_z(0^+, t). \quad (\text{A.7})$$

Outlet ($z = L$): $c_z(L, t) = 0$, $T_z(L, t) = 0$ (no dispersive flux through the exit plane).

Danckwerts is a Robin condition; it makes the inventory identity of Part B exact and forces $c(0^+, t) < c_f$ while the front is inside the bed (the dispersive term is positive). Dirichlet overfeeds the column by $-\varepsilon D_L c_z(0, t) > 0$; the defect is $O(Pe^{-1})$ and vanishes in the hyperbolic limit. Both are implemented in Part E; the finite-volume form makes Danckwerts trivial.

A.6 Full model (summary)

$$\begin{aligned} \varepsilon c_t + uc_z &= \varepsilon D_L c_{zz} - \alpha_b q_t, & q_t &= k(q^*(c, T) - q), \\ C_h T_t + u \rho_g c_{p,g} T_z &= \lambda_{\text{eff}} T_{zz} + \alpha_b (-\Delta H) q_t - \frac{4h_w}{d_{\text{col}}} (T - T_{\text{wall}}), \end{aligned}$$

with closure (??) or (??), IC (??), BC (??) + zero-gradient outlet. Structurally this is a **parabolic–hyperbolic relaxation system**: two transport PDEs coupled pointwise to one stiff local ODE.